Quasi-Poisson models

Last time: modeling article publication

We are interested in analyzing the number of articles published by biochemistry PhD students. The data contains the following variables:

- `art`: articles published in last three years of Ph.D.
- `fem`: gender (recorded as male or female)
- `mar`: marital status (recorded as married or single)
- `kid5`: number of children under age six
- `phd`: prestige of Ph.D. program
- `ment`: articles published by their mentor in last three years

$$Articles_i \sim Poisson(\lambda_i)$$

$$\log(\lambda_i) = \beta_0 + \beta_1 Female_i + \beta_2 Married_i + \beta_3 Kids_i + \beta_4 Prestige_i + \beta_5 Mentor_i$$

Recap: confidence interval

	Estimate	Std. Error	z value	Pr(>|z|)	
(Intercept)	0.304617	0.102981	2.958	0.0031	**
femWomen	-0.224594	0.054613	-4.112	3.92e-05	***
marMarried	0.155243	0.061374	2.529	0.0114	*
kid5	-0.184883	0.040127	-4.607	4.08e-06	***
phd	0.012823	0.026397	0.486	0.6271	
ment	0.025543	0.002006	12.733	< 2e-16	***

Confidence interval for β_4:

95% : $0.0128 \pm 1.96\,(0.0264)$

Goodness of fit test

```
    Null deviance: 1817.4  on 914  degrees of freedom
Residual deviance: 1634.4  on 909  degrees of freedom
```

H_0: model is a good fit to the data

H_A: model is not a good fit to the data

Test statistic: residual deviance $= 1634.4$

Under H_0: residual deviance $\sim \chi^2_{\text{residual df}}$

$\Rightarrow \chi^2_{909}$

pchisq(1634.4, 909, lower.tail = F) ≈ 0

Model does not seem to be a good fit

Offest?

- `art`: articles published in last three years of Ph.D.
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$$Articles_i \sim Poisson(\lambda_i)$$

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> Do you think this model needs an offset?

Overdispersion

Overdispersion occurs when the response Y has higher variance than we would expect if Y followed a Poisson distribution.

Formally, let

dispersion parameter

$$\phi = \frac{\text{Variance}}{\text{Mean}}$$

(specific to a Poisson)

> What should ϕ be if there is no overdispersion?

If Poisson is correct, then variance = mean

$\phi = 1$

Estimating overdispersion

The *Pearson residual* for observation i is defined as

$$e_{(P)i} = \frac{Y_i - \widehat{\lambda}_i}{\sqrt{\widehat{\lambda}_i}}$$

difference between observed and predicted value

std deviation of y

$$\widehat{\phi} = \frac{\sum_{i=1}^{n} e^2_{(P)i}}{n - p}$$

residual degrees of freedom

- $p =$ number of parameters in model

Example: Estimating overdispersion

```
# fit the model
m1 <- glm(art ~ ., data = articles,
          family = poisson)

# get Pearson residuals
pearson_resids <- resid(m1, "pearson")

# estimate dispersion parameter
phihat <- sum(pearson_resids^2)/(915 - 6)
phihat
```

```
## [1] 1.828984
```

Variance is about 1.83 times mean

Handling overdispersion

Overdispersion is a problem because our standard errors (for confidence intervals and hypothesis tests) are too low.

> If we think there is overdispersion, what should we do?

- option 1: use $\hat{\phi}$ to adjust SEs in fitted model
- option 2: use a different model / distribution (e.g., negative binomial)

Adjusting the standard error

- In our data, $\widehat{\phi} = 1.83$
- This means our variance is 1.83 times bigger than it should be
- So our standard error is $\sqrt{1.83} = 1.35$ times bigger than it should be

$$SE(\hat{\beta})_{\text{overdispersion}} = \sqrt{\phi} \cdot SE(\hat{\beta})_{\text{Poisson}}$$

Adjusting the standard error in R

```
m2 <- glm(art ~ ., data = articles,
          family = quasipoisson)
```

estimated coefficients are the same!

$0.0357 = \sqrt{1.83} \cdot 0.0264$

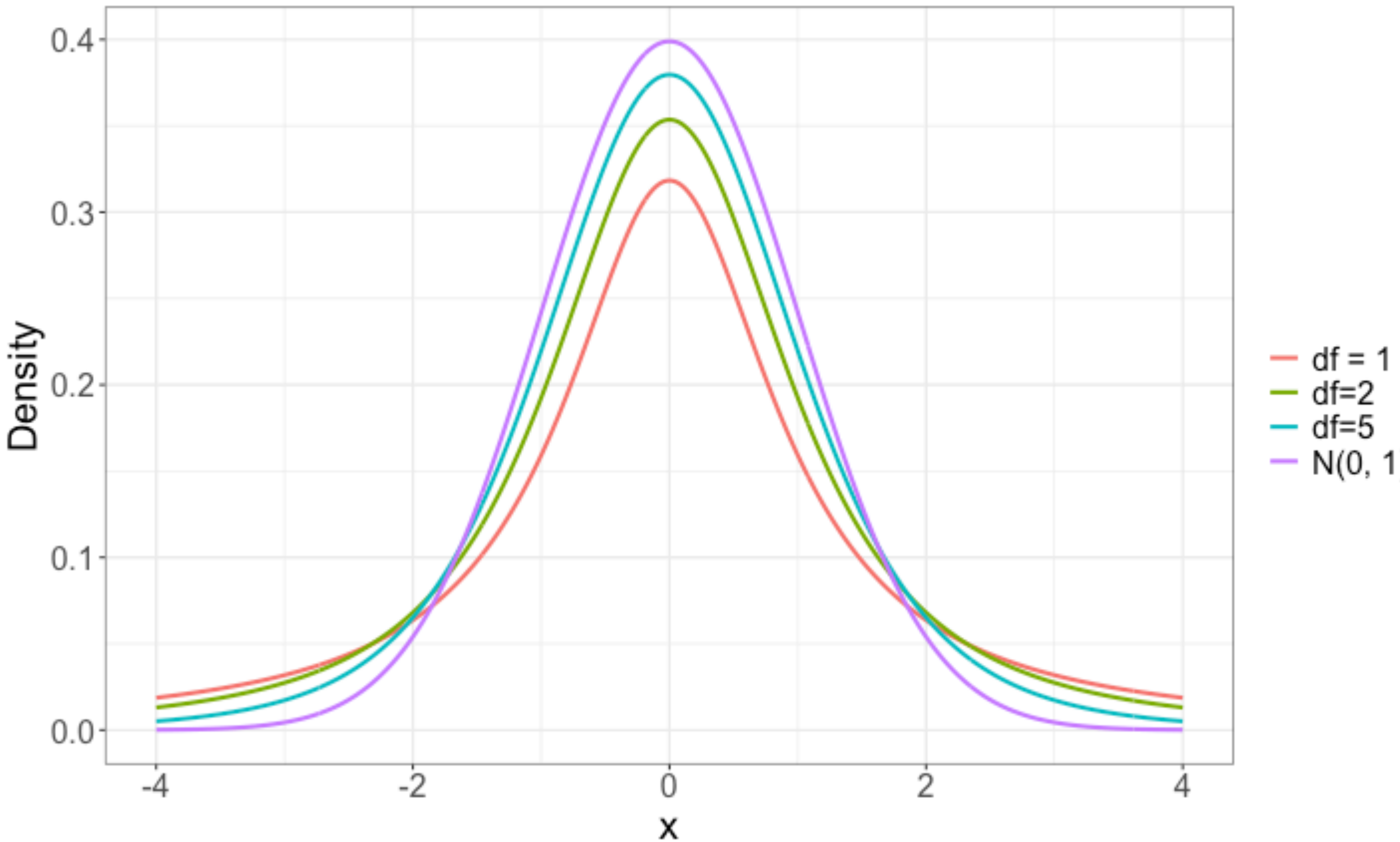
```
...
## kid5   -0.184883   0.054268  -3.407 0.000686 ***
## phd     0.012823   0.035700   0.359 0.719544
## ment    0.025543   0.002713   9.415  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.829006)
##
##     Null deviance: 1817.4  on 914  degrees of freedom
## Residual deviance: 1634.4  on 909  degrees of freedom
## AIC: NA
...
```

$\hat{\phi} = 1.83$

- Allowing ϕ to be different from 1 means we are using a *quasi-likelihood* (in this case, a *quasi-Poisson*)

Intuition: Estimating ø => more variability in test statistics & confidence intervals

t-distribution



Calculating a confidence interval

```
...
## phd          0.012823   0.035700   0.359 0.719544
## ment         0.025543   0.002713   9.415  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasipoisson family taken to be 1.829006)
...
```

New confidence interval for β_4:

t^*_{n-p} = critical value for a t distribution w/ n-p df

$$0.0128 \pm t^*_{n-p} \cdot 0.0357$$

```
qt(0.025, df=909, lower.tail=F)
```

```
## [1] 1.962577
```

$$0.0128 \pm 1.96 \cdot 0.0357 = (-0.0572,\ 0.0828)$$

Adjusting the standard error in R

Poisson:

```
...
## kid5          -0.184883   0.040127  -4.607 4.08e-06 ***
## phd            0.012823   0.026397   0.486   0.6271
## ment           0.025543   0.002006  12.733  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
...
```

Quasi-Poisson:

```
...
## kid5          -0.184883   0.054268  -3.407 0.000686 ***
## phd            0.012823   0.035700   0.359 0.719544
## ment           0.025543   0.002713   9.415  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1
...
```

Class activity

Work on the class activity. Submit your rendered HTML on Canvas at the end of class.